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Methods, systems, and media for identifying human coactivity in images and videos using neural networks

Abstract

Methods, systems and processor-readable media for classifying human coactivity performed jointly by two humans shown in an image or a sequence of frames of a video. A 2D convolutional neural network is used to identify key points on the human body, such as human body joints, visible within the image or within each frame, for each of the two people performing the coactivity. An encoded representation of the key points is created for each image or frame, the encoded representation being based on distances between the key points of the first person and key points of the second person. The encoded representation for the image, or a concatenated volume of the encoded representations of the frames, is processed by a fully-connected neural network trained to classify the coactivity.

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Background/Summary

RELATED APPLICATION DATA

(1) This is the first patent application related to this matter.

TECHNICAL FIELD

(2) The present disclosure relates to human body tracking in a digital video, and in particular to methods, systems, and processor readable media for identifying a coactivity of two human bodies in an image or a digital video.

BACKGROUND

(3) Recognizing human behavior is a longstanding problem in computer vision research. While

machine learning has resulted in advances in the realm of computer vision, including a range of approaches to identifying humans in digital images or digital videos, and for tracking the movement of human bodies over multiple frames of a digital video, the recognition or identification of specific types of human behavior in a digital video remains difficult.

(4) Most existing approaches to human behavior recognition attempt to classify activities performed by a single person, such as walking, jogging, and jumping. Many such existing approaches use machine learning to train neural networks to perform 3D convolution operations: one such example is described by S. Ji, W. Xu, M. Yang and K. Yu, "3D Convolutional Neural Networks for Human Action Recognition," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 35, no. 1, pp. 221-231, January 2013, doi: 10.1109/TPAMI.2012.59. Such 3D convolution-based approaches are limited to classification of activities performed by a single person in isolation.

(5) However, many activities are performed by more than one person, such that the interaction between two or more people is what defines the activity. Examples of two-person activities, also referred to herein as "coactivities", include partner dance or a foul play in a soccer match. The information necessary to identify or classify such coactivities is unusable by the existing approaches to identification of single-person activities described above, and those existing approaches are incapable of classifying coactivities that are defined by the interaction of two or more human bodies instead of by independent action by each of two or more human bodies.

(6) Accordingly, it would be desirable to provide a method for classifying coactivity by two or more human bodies visible in an image or a time series of images (e.g., a sequence of video frames).

SUMMARY

(7) The present disclosure provides methods, systems, and processor readable media for identifying a coactivity of two human bodies in an image or a digital video. Each image (e.g., each video frame) is processed to extract key point data, each key point corresponding to a predefined key point on a human body visible in the image: e.g., a set of 18 key points may be extracted for each visible human body, each of the 18 key points corresponding to a predefined location on the body, such as the left knee, the right eye, and so on. A first key point position set is generated, representative of the position of each key point of a first person. A second key point position set is generated, representative of the position of each key point of a second person. Optionally, one or more key points of an object visible in the image may also be extracted, along with its position.

(8) An encoded representation of the image is generated by measuring a respective 2D or 3D distance from each key point of the first human body to each key point of the second human body. In the example above in which each body has 18 key points, the encoded representation includes $(18*18)=324$ distance values. Optionally, the encoded representation may also include a distance from one or more key points of each human body to the key point(s) of the object. In the case of video data (i.e. a plurality of images), the encoded representation of each image (i.e., each video frame) may be concatenated into a 3D data volume of size $(18*18*n)$, wherein n is the number of video frames.

(9) The encoded representation of the image(s) is provided to a coactivity classifier to generate classification information representative of a coactivity identified in the image(s). The coactivity classifier may include a model, such as a fully-connected neural network, trained using machine learning to classify coactivity based on encoded representations of images.

(10) In at least some aspects, the disclosure relates to a method. For each image of one or more images, a number of steps are performed. A first key point position set and a second key point position set for the image are obtained. The first key point position set includes a key point position for each key point of a plurality of key points of a first human body detected in the image, and the second key point position set includes a key point position for each key point of a plurality of key points of a second human body detected in the image. The first key point position set and the

second key point position set are processed to determine a distance between each key point position of the first key point position set and each key point position of the second key point position set. The distances between each key point position of the first key point position set and each key point position of the second key point position set are processed to generate an encoded representation of the image. The encoded representations of each image of the one or more images are provided to a coactivity classifier that includes a machine learned model that is configured to classify a coactivity of two human bodies. Classification information is generated, using the coactivity classifier, classifying a coactivity performed by the first human body and second human body in the one or more images.

(11) In at least some aspects, the disclosure relates to a system, comprising a processor and a memory having stored thereon executable instructions. The executable instructions, when executed by the processor, cause the system to perform a number of operations. For each image of one or more images, a number of steps are performed. A first key point position set and a second key point position set for the image are obtained. The first key point position set includes a key point position for each key point of a plurality of key points of a first human body detected in the image, and the second key point position set includes a key point position for each key point of a plurality of key points of a second human body detected in the image. The first key point position set and the second key point position set are processed to determine a distance between each key point position of the first key point position set and each key point position of the second key point position set. The distances between each key point position of the first key point position set and each key point position of the second key point position set are processed to generate an encoded representation of the image. The encoded representations of each image of the one or more images are provided to a coactivity classifier that includes a machine learned model that is configured to classify a coactivity of two human bodies. Classification information is generated, using the coactivity classifier, classifying a coactivity performed by the first human body and second human body in the one or more images.

(12) In at least some aspects, the disclosure relates to a non-transitory processor-readable medium containing instructions which, when executed by a processor of a system, cause the system to perform a number of operations. For each image of one or more images, a number of steps are performed. A first key point position set and a second key point position set for the image are obtained. The first key point position set includes a key point position for each key point of a plurality of key points of a first human body detected in the image, and the second key point position set includes a key point position for each key point of a plurality of key points of a second human body detected in the image. The first key point position set and the second key point position set are processed to determine a distance between each key point position of the first key point position set and each key point position of the second key point position set. The distances between each key point position of the first key point position set and each key point position of the second key point position set are processed to generate an encoded representation of the image. The encoded representations of each image of the one or more images are provided to a coactivity classifier that includes a machine learned model that is configured to classify a coactivity of two human bodies. Classification information is generated, using the coactivity classifier, classifying a coactivity performed by the first human body and second human body in the one or more images.

(13) In some examples, the one or more images comprises a plurality of frames of a video. Providing the encoded representation of each image of the one or more images to the coactivity classifier comprises concatenating the encoded representations of the plurality of frames to generate a three-dimensional (3D) data volume, and providing the 3D data volume as input to the coactivity classifier.

(14) In some examples, the one or more images comprises a single image. The encoded representation of each image of the one or more images comprises a vector of values. Each value is based on the distance between each key point position of the first key point position set and each

key point position of the second key point position set.

(15) In some examples, each key point position is a two-dimensional (2D) position within a plane of the respective image, and each distance is a 2D distance within an image plane of the respective image.

(16) In some examples, each key point position is a three-dimensional (3D) position representative of a location of the respective key point in 3D space, and each distance is a 3D distance.

(17) In some examples, further steps are performed for each image of the one or more images. An object key point position for the image is obtained. The object key point position is representative of a position of a key point of an object detected in the image. The first key point position set and the object key point position are processed to determine a distance between each at least one key point position of the first key point position set and the object key point position, and the second key point position set and the object key point position are processed to determine a distance between each at least one key point position of the second key point position set and the object key point position. For each image of the one or more images, the encoded representation of the image is further generated by processing the distance between each at least one key point position of the first key point position set and the object key point position, and the distance between each at least one key point position of the second key point position set and the object key point position.

(18) In some examples, obtaining the first key point position set and the second key point position set for the image comprises generating the first key point position set and the second key point position set using a pose estimation model. The pose estimation model is configured to perform a number of steps. The pose estimation model receives a first bounding box for the first human body, comprising one or more pixel values of a first plurality of pixels of the respective image. The pose estimation model receives a second bounding box for the second human body, comprising one or more pixel values of a second plurality of pixels of the respective image. The pose estimation model processes the first bounding box to generate the key point position for each key point of the plurality of key points of the first human body within the first bounding box, and processes the second bounding box to generate the key point position for each key point of the plurality of key points of the second human body within the second bounding box.

(19) In some examples, the machine learned model comprises a fully-connected neural network.

(20) In some examples, at least one key point position of the first key point position set corresponds to a joint of the first human body, and at least one key point position of the second key point position set corresponds to a joint of the second human body.

(21) In some examples, the encoded representation of each respective image comprises a vector comprising a plurality of distance values. The plurality of distance values comprises the distance between each key point position of the first key point position set and each key point position of the second key point position set of the respective image.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will now be described by way of examples with reference to the accompanying drawings, in which like reference numerals may be used to indicate similar features.

(2) FIG. 1A is a schematic diagram showing operations of a first example system for identifying human coactivity in a single image, according to example embodiments described herein.

(3) FIG. 1B is a schematic diagram showing operations of a second example system for identifying human coactivity in a sequence of video frames, according to example embodiments described herein.

(4) FIG. 2A is a schematic diagram showing further details of the encoded representation generation and coactivity classification blocks of the first example system of FIG. 1A.

- (5) FIG. 2B is a schematic diagram showing further details of the encoded representation generation and coactivity classification blocks of the second example system of FIG. 1B.
- (6) FIG. 3 is a block diagram of an example computing system for classifying coactivity in one or more images according to the first example system of FIG. 1A or the second example system of FIG. 1B.
- (7) FIG. 4A is an example of an image showing key points superimposed on an image of a human body by example embodiments described herein.
- (8) FIG. 4B is a visual representation of a plurality of key point coordinates as identified in the example image shown in FIG. 4A by example embodiments described herein.
- (9) FIG. 5 is an example of an image showing key points superimposed on an image of three human bodies by example embodiments described herein, wherein two of the human bodies are engaged in a coactivity.
- (10) FIG. 6 shows example data representations of the first key point position set, second key point position set, and encoded representation of FIG. 2A.
- (11) FIG. 7 is a flowchart of a first example method for classifying human coactivity in one or more images according to example embodiments described herein.
- (12) FIG. 8 is a flowchart of a second example method for classifying human coactivity involving an object in one or more images according to example embodiments described herein.
- (13) FIG. 9 is a flowchart of an example method for obtaining the first key point position set and second key point position set according to the first step of the method of FIG. 7.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(14) The present disclosure is made with reference to the accompanying drawings, in which embodiments are shown. However, many different embodiments may be used, and thus the description should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same elements, and prime notation is used to indicate similar elements, operations or steps in alternative embodiments. Separate boxes or illustrated separation of functional elements of illustrated systems and devices does not necessarily require physical separation of such functions, as communication between such elements may occur by way of messaging, function calls, shared memory space, and so on, without any such physical separation. As such, functions need not be implemented in physically or logically separated platforms, although they are illustrated separately for ease of explanation herein. Different devices may have different designs, such that although some devices implement some functions in fixed function hardware, other devices may implement such functions in a programmable processor with code obtained from a machine-readable medium. Lastly, elements referred to in the singular may be plural and vice versa, except where indicated otherwise either explicitly or inherently by context.

(15) Example embodiments will now be described with respect to methods, systems, and non-transitory media for identifying a coactivity of two human bodies in an image or a digital video.

(16) FIG. 1A is a schematic diagram showing the operations of a first example system **10** for classification of human coactivity in a single image **12**.

(17) In some embodiments, the image **12** is a two-dimensional (2D) image that includes a corresponding RGB (red, green, and blue) value for each pixel in the 2D image. The pixel color values may be mapped to a 2D x-y coordinate system wherein x is a horizontal pixel position beginning at 0 at the left edge of the image and increasing toward the right edge, and y is a vertical pixel position beginning at 0 at the top edge of the image and increasing toward the bottom edge. In some embodiments, the image **12** includes a corresponding RGB (red, green, and blue) value for each pixel in the image **12** as well as a depth (z) value for each pixel in the image **12**, wherein the depth value indicates a pixel position beginning at 0 at the position of the camera used to capture the image and increasing away from the camera into the scene. A camera having depth-finding

capabilities may capture the image **12** and generate the depth (z) value for each pixel in the image **12**. Alternatively, pre-processing of the image **12** may be performed to extract a depth value for each pixel corresponding to a human detected in the image **12**.

(18) Example embodiments will be described below with reference to 2D images without depth information, in which the position of a key point extracted from the image **12** is a 2D position (e.g., x and y pixel coordinates), and in which a distance between two key points is a 2D distance within the image plane of the image (e.g., within the 2D pixel space of the image defined by x and y pixel coordinates). However, it will be appreciated that embodiments using depth information as part of the image **12** may represent key point positions as 3D positions (e.g., an x pixel coordinate, a y pixel coordinate, and a z depth value for each key point), and may calculate and represent a distance between two key points as a 3D distance.

(19) The system **10** begins by processing the image **12** at a first bounding box generation block **14** to generate a first bounding box corresponding to a first human body visible in the image **12**, and processing the image **12** at a second bounding box generation block **16** to generate a second bounding box corresponding to a second human body visible in the image **12**. Each bounding box may substantially surround all or part of its respective human body. An example bounding box generated in association with a human body detected in an image is described with reference to FIG. **4A** below. Bounding box generation based on object detection, and specifically based on detection of human bodies, is a well known problem in computer vision, and any suitable bounding box generation technique may be used by first bounding box generation block **14** and second bounding box generation block **16**. It will be appreciated that first bounding box generation block **14** and second bounding box generation block **16** may both be implemented by the same logic in some embodiments, e.g., a single convolutional neural network trained to perform object detection using reinforcement learning.

(20) The first bounding box, and optionally the image **12**, are processed by a first pose estimation block **18** to generate the first key point position set **20**. The second bounding box, and optionally the image **12**, are processed by a second pose estimation block **19** to generate the second key point position set **22**. It will be appreciated that first pose estimation block **18** and second pose estimation block **19** may both be implemented by the same logic in some embodiments. In some embodiments, the first pose estimation block **18** and second pose estimation block **19** are implemented by a two-dimensional (2D) convolutional neural network (CNN) that has been trained to identify key points in images as described in further detail below. The trained 2D CNN performs feature extraction to identify a set of key points for a respective human body detected in the image **12**, as indicated by the respective bounding box generated in association with the respective human body. The trained 2D CNN may be a trained body tracking or pose estimation CNN. An example of a trained pose estimation CNN is described by Alexander Toshev and Christian Szegedy in *DeepPose: Human Pose Estimation via Deep Neural Networks*, arXiv:1312.4659, <https://arxiv.org/abs/1312.4659>, which is hereby incorporated by reference as if reproduced in their entirety. Another example of a trained pose estimation CNN is described in Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, and Yaser Sheikh in *OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields*, arXiv: 1812.08008, <http://arxiv.org/abs/1812.08008>, which is also hereby incorporated by reference as if reproduced in their entirety.

(21) Each key point corresponds to a position of a pixel in the image **12**. Each key point includes a horizontal coordinate (x) and vertical coordinate (y) for the position of the pixel in the image **12** that has been identified as a key point. As described above, embodiments making use of depth data may also include a depth coordinate (z) for the position of the pixel in the image **12** that has been identified as a key point.

(22) In some embodiments, the key points correspond to the positions in the frame of various joints or other locations on a human (such as eyes, ears, nose, and neck) detected and tracked in the image **12**. An example configuration of correspondences between key points and locations on a

human body is described below with reference to FIG. 4B.

(23) The first key point position set **20** is representative of the position of each key point of the first human body in the image **12**, and the second key point position set **22** is representative of the position of each key point of the second human body in the image **12**. An encoded representation generation block **23** processes the first key point position set **20** and second key point position set **22** to generate an encoded representation **24** of the image **12**. The encoded representation **24** is generated and encoded as described with reference to FIG. 6 below. In some embodiments, the encoded representation **24** is generated by computing a distance value between each key point of the first person (i.e. the first key point position set **20**) and each key point of the second person (i.e. the second key point position set **22**). The distance value may be computed as a 2D distance for images having key points defined in a 2D coordinate space as described above, as a 3D distance for images having key points defined in a 3D coordinate space as described above, or as another type of distance value representative of a distance between a given pair of key points. In some embodiments, the encoded representation **24** may omit distance values for one or more of the key points: for example, only key points corresponding to joints of the human body may be used to compute distance values in some embodiments (such as embodiments configured to classify coactivities that do not involve user's faces).

(24) Finally, the encoded representation **24** of the image **12** is processed by a coactivity classifier **26** to generate classification information, as described in greater detail below with reference to FIG. 2A.

(25) FIG. 1B is a schematic diagram showing the operations of a second example system **50** for classification of human coactivity in a sequence of video frames (shown as frame [1] **52A** through frame [n] **52B**). The second example system **50** processes each frame **52A** through **52B** using the same functional blocks **14**, **16**, **18**, **19** to generate a first key point position set **20** and second key point position set **22** for each frame. However, the encoded representation generation block **53** of the second example system **50** may differ from the encoded representation generation block **23** of the first example system **10** insofar as each encoded representation generation block **53** operates to concatenate or otherwise combine the encoded representations of the sequence of frames **52A** through **52B** into a combined encoded representation **54** of frames [1] through [n], as described in greater detail below with reference to FIG. 2B. The coactivity classifier **56** of the second example system **50** is configured to receive the combined encoded representation **54** as input and to classify coactivity shown in the sequence of frames **52A** through **52B**, as represented by the combined encoded representation **54**, instead of classifying coactivity shown in a single image as represented by the encoded representation **24** of FIG. 1A.

(26) FIG. 2A is a schematic diagram showing further details of example encoded representation generation **23** and coactivity classification **26** operations of the first example system **10**. In this example, the first key point position set **20** includes 18 key point positions, denoted as key point positions K1(0) through K1(17): for example, the set of key points identified for each human body in the image **12** may be the 18 key points described below with reference to FIG. 4B. Similarly, the second key point position set **21** includes 18 key point positions, denoted as key point positions K2(0) through K2(17).

(27) In this example, the encoded representation generation block **23** generates the encoded representation **24** of the image **12** by computing a distance between each K1 and each K2 (i.e., between each key point position K1 of the first key point position set **20** and each key point position K2 of the second key point position set **22**), shown as distance computation operation **202**. The distance between two key point positions in a 2D space may be computed as $\text{distance}(K1, K2) = \sqrt{(x_{\text{sub.2}} - x_{\text{sub.1}})^2 + (y_{\text{sub.2}} - y_{\text{sub.1}})^2}$ wherein $K1 = (x_{\text{sub.1}}, y_{\text{sub.1}})$ and $K2 = (x_{\text{sub.2}}, y_{\text{sub.2}})$. Similarly, the distance between two key point positions in a 3D space may be computed as $\text{distance}(K1, K2) = \sqrt{(x_{\text{sub.2}} - x_{\text{sub.1}})^2 + (y_{\text{sub.2}} - y_{\text{sub.1}})^2 + (z_{\text{sub.2}} - z_{\text{sub.1}})^2}$ wherein $K1 = (x_{\text{sub.1}}, y_{\text{sub.1}}, z_{\text{sub.1}})$ and $K2 =$

(x.sub.2, y.sub.2, z.sub.2).

(28) The distance computation operation **202** generates a set of distance values, wherein the number of distance values in the set is equal to the square of the number of key points of a human body: in this example, using 18 key points per human body, the number of distance values in the set is $(18*18)=324$ distance values. The **324** distance values may be encoded as a 324-element distance vector **204**. The encoded representation **24** of the image **12** may include the distance vector **204**, by itself or in combination with other information.

(29) The coactivity classifier **26** in this example includes a feature extraction neural network **206** and a coactivity classification neural network **208**. These two neural networks **206**, **208** interact to implement a model, trained using machine learning, for classifying coactivity based on the encoded representation **24**. First, the feature extraction neural network **206** processes the encoded representation **24** as input, propagating the encoded representation **24** forward through one or more layers of the feature extraction neural network **206** to generate a set of one or more feature maps or activation maps as output. The output of the feature extraction neural network **206** is then processed by the coactivity classification neural network **208** to generate classification information **210** classifying a coactivity performed by the first human body and the second human body in the image **12**.

(30) In some embodiments, the feature extraction neural network **206** is a fully-connected neural network, and the coactivity classification neural network **208** is a single neural network output layer including a softmax function for normalizing a set of logits generated by the neural network output layer to generate the classification information **210** as a probability distribution across a set of coactivity classes. In some embodiments, the coactivity class predicted with the highest probability in the normalized probability distribution may be used as the final output of the system **10**.

(31) The set of coactivity classes may be defined based on the intended problem domain: for example, a system configured to identify foul plays from images of soccer games may use the set of coactivity classes {"foul", "not foul"}, whereas a system configured to identify types of partner dance from images of ballroom dancing may use the set of coactivity classes {"tango", "salsa", "rumba", "waltz", "foxtrot", "not dancing"}. In some embodiments, the feature extraction neural network **206** and coactivity classification neural network **208** are jointly trained, using reinforcement learning, with a training dataset containing images labelled with one of the coactivity classes in the set.

(32) FIG. 2B is a schematic diagram showing further details of example encoded representation generation **53** and coactivity classification **56** operations of the second example system **50**. The pair of key point positions sets **20**, **22** for each image (i.e. each frame in the sequence of frames [1] **52A** through [n] **52B**) is processed by the encoded representation generation block **53** by performing the same distance computation operation **202** as in FIG. 2A. However, each encoded representation generation block **53** (or each iteration of the encoded representation generation block **53** applied to a separate frame) provides the output of the distance computation operation **202** to a concatenation block **252**. The concatenation block **252** concatenates the n distance vectors corresponding to the n frames **52A** through **52B** to generate a 3D data volume **254** of dimensions (# of key points in first key point position set*# of key points in second key point position set*n), i.e., in this example, $(18*18*n)=(324*n)$ distance values.

(33) The 3D data volume **254** is then provided as input to the coactivity classifier **56**. In some embodiments, the 3D data volume **254** is flattened into a linear vector of $(324*n)$ elements before being processed by the coactivity classifier **56**. As in the coactivity classifier **26** of FIG. 2A, the coactivity classifier **56** includes a feature extraction neural network **256** and a coactivity classification neural network **258**; in the multi-frame context of FIG. 2B, these networks **256**, **258** are trained and configured to classify coactivities being performed by two human bodies over a sequence of n video frames, using a training data set containing labelled video clips of n frames

each.

(34) In some embodiments, the feature extraction neural network **256** is a convolution neural network configured to generate as output a final feature vector of 256 elements. The final feature vector is provided as input to the coactivity classification neural network **258**. In some embodiments, the coactivity classification neural network **258** is a fully-connected neural network that has a final output layer for classifying the coactivity using a softmax function, as described above with reference to the final layer of the coactivity classification neural network **208**. The fully-connected neural network of the coactivity classification neural network **258** includes one or more hidden layers, each having a relu activation function, followed by the final output layer having the softmax activation function for generating the classification information **210**.

(35) In some embodiments, a system **10** or **50** may be configured to further identify and encode additional key points detected within the image **12** or within one or more of the frames **52A** through **52B**. For example, one or more key points of an additional object detected within an image **12** may also be processed to generate the encoded representation **24**. Examples of such additional object key points and their inclusion in the operations described above will be described below with reference to FIGS. **5** and **8**.

(36) FIG. **3** shows a block diagram of a computing system **300** for implementing one or more of the example systems **10**, **50** described above with reference to FIGS. **1A-2B**. The computing system **300** includes a processor **302** for executing computer program instructions, and a memory **304** for storing executable instructions and data.

(37) The processor **302** may be embodied as any processing resource capable of executing computer program instructions, such as one or more processors on a computer or computing platform(s). The memory **304** may be embodied as any data storage resource, such as one or more disk drives, random access memory, or volatile or non-volatile memory on one or more computing platforms. It will be appreciated that, in some embodiments, the computing system **300** may be a distributed computing system implemented over two or more separate hardware platforms in remote communication with each other.

(38) The memory **304** has stored thereon several types of computer programs in the form of executable instructions. It has thereon a set of executable instructions **310** for one or more of the systems, such as example systems **10** and **50**, configured to perform the methods described herein. It also has stored thereon one or more sets of instructions of trained neural networks or other machine learned models to identify key points in frames (e.g., a trained CNN used by the pose estimation blocks **18** and **19**) and/or to classify human coactivity based on encoded representations (e.g., feature extraction neural network **206** and coactivity classification neural network **208**), shown here as pose estimation neural network instructions **322**, feature extraction neural network instructions **324**, and coactivity classification neural network instructions **326** respectively.

(39) The memory **304** may have stored thereon several types of data **380**. The data **380** may include image data **301** representative of the image **12** or of the sequence of frames **52A** through **52B**. In some embodiments, the image data **301** may be received from an external source via a communication system (not shown), potentially being stored in part in a buffer or cache in the memory **304**. The memory **304** may also have stored thereon the classification information **210** generated by the system **10**, **50**. The memory **304** may also store, temporarily or otherwise, the various types of intermediate information or data (not shown) used by the system **10**, **50**, such as the encoded representations **24** and/or **54**, the first and second key points position sets **20**, **22**, etc.

(40) FIG. **4A** shows an example image **12** that includes a bounding box **416** generated for a human body **412** detected in the image **12** and the key points identified for the human body overlaid on the image. The image **12** shows the human body **412** against a background **413**. Key points identified in the image **12** by the pose estimation block **18**, **19** are shown as dots, such as a right hip key point **424**, a left elbow key point **422**, and a left wrist key point **420**. In the image **12** shown in FIG. **4A**, the bounding box **416** is shown as a rectangular box overlaid on the image **12** that encompasses the

torso of the human body **412** i.e. the shoulder key points, neck key point, and hip key points.

(41) FIG. **4B** shows an example key point skeleton **440** with identified key points corresponding to positions on a human body. In this example, each key point on the skeleton **440** is assigned a unique identifier in the form of an index number, from #1 to #17. Most of the key points (e.g. key points assigned index numbers #0 through #13) correspond to joints of the skeleton **440**, while a few key points (key points assigned index numbers #0 and #14 through #17) correspond to fixed locations on the head, such as eyes, nose, and ears. For example, index number #8 corresponds to the right hip, and is shown in FIG. **4A** as right hip key point **424** and right hip key point **474** in FIG. **4B**. Index number #7 corresponds to the left wrist, and is shown as left wrist key point **420** in FIG. **4A** and left wrist key point **470** in FIG. **4B**. Left wrist key point **570** in FIG. **4B** is marked with an index number “7”. Index number #6 corresponds to the left elbow, and is shown as left elbow key point **422** in FIG. **4A** and left elbow key point **472** in FIG. **4B**. Left elbow key point **472** is marked with the index number “6”. Various body parts may be identified on the skeleton **440** as the segments extending between two adjacent joints, such as left lower arm **480** between the left wrist key point **470** corresponding to index number #7 and the left elbow key point **472** corresponding to index number #6.

(42) The key points shown in FIG. **4B** are listed in the table below with their respective index numbers:

(43) TABLE-US-00001

Index Number	Body Location
#0	Nose
#1	Neck
#2	Right Shoulder
#3	Right Elbow
#4	Right Wrist
#5	Left Shoulder
#6	Left Elbow
#7	Left Wrist
#8	Right Hip
#9	Right Knee
#10	Right Ankle
#11	Left Hip
#12	Left Knee
#13	Left Ankle
#14	Right Eye
#15	Left Eye
#16	Right Ear
#17	Left Ear

(44) A key point position includes (x,y) or (x,y,z) pixel coordinates of a key point with respect to the coordinate system of a frame. Thus, assuming the lower left corner of the image **12** has pixel coordinates (0,0), the key point **474** corresponding to index number #8 (shown as right hip key point **424** in FIG. **4A**) might have pixel coordinates (100,80), and the key point position for key point **472** corresponding to index number #6 (shown as left elbow key point **422** in FIG. **4A**), might have pixel coordinates (190,170). As the knees and ankles of the body **412** are not visible within the image **12**, the knee key points (corresponding to index numbers #9 and #12) and the ankle key points (corresponding to index numbers #10 and #13) are not identified by the pose estimation block **18**, **19** and are not shown as dots overlaid within the image **12**.

(45) FIG. **5** shows an example image **500** in which three human bodies **502**, **504**, **506** and a soccer ball **508** are visible. The image **500** shows key points identified for each of the human bodies **502**, **504**, **506** and the soccer ball **508**. Bounding boxes are not shown, but in some embodiments may be generated prior to identifying the key point positions, as described above. For example, the image may be processed by an object detection model to generate bounding boxes for each human body **502**, **504**, **506** and the soccer ball **508** prior to identifying the key points shown in the image **500**.

(46) Example systems and methods described herein may be used to process key points of the first human body **502** and second human body **504** to classify a human coactivity being jointly performed thereby. For example, system **10** may use a coactivity classifier **26** trained to generate classification information **210** classifying the coactivity of the first and second human bodies **502**, **504** among a set of coactivity classes {“foul”, “not foul”} indicating whether or not a foul play is being performed in the depicted soccer game. In some embodiments, the third human body **506** may be omitted from processing by the system **10**, for example based on the relative proximities of each of the human bodies **502**, **504**, **506**. In other embodiments, each pair of human bodies (**502** and **504**, **502** and **506**, **504** and **506**) may be provided separately to the system **10** for coactivity classification.

(47) The first key point position set **20** may be generated with respect to the first human body **502**, encompassing all of the visible key points of the first human body **502**, such as left wrist key point **514** and left knee key point **512**. The second key point position set **22** may be generated with

respect to the second human body **504**, encompassing all of the visible key points of the second human body **504**, such as right elbow key point **522** and right hip key point **524**.

(48) In some examples, further described below with reference to the method of FIG. **8**, an object in the image **500** (such as the soccer ball **508**) may be identified and assigned one or more key points, for example by an object detection model. In image **500**, the soccer ball **508** has a single key point **542**, located approximately at the center of the soccer ball **508**.

(49) FIG. **6** shows an example of how the first key point position set **20** and second key point position set **22** are processed using the distance computation operation **202** to generate the distance vector **204** of the encoded representation **24**. The first key point position set **20** and second key point position set **22** each include 18 key points positions, denoted by their respective key point identifiers (i.e. index numbers 0 through 17, as described above, denoted as K1(0) through K1(17) for the first key point position set **20** and as K2(0) through K2(17) for the second key point position set **22**) and represented as 2D coordinate sets. For example, key point position K1(15) has position K1(15)=(x15,y15), whereas key point position K2(7) has position K2(7)=(x7',y7'). It will be appreciated that the encodings shown in FIG. **6** can be extended to 3D coordinates as described above.

(50) As described above, the distance computation operation **202** computes a distance between each key point of the first key point position set **20** and each key point of the second key point position set **22** to generate the distance vector **204**. Thus, the first element of the distance vector **204**, denoted as Key Point Pair (0,0), is shown as a distance value computed between key point K1(0) (i.e., the nose key point, having index number 0, from the first key point position set **20**) and key point K2(0) (i.e., the nose key point, having index number 0, from the second key point position set **22**). The distance value of this first element is computed as $\sqrt{(x_0' - x_0)^2 + (y_0' - y_0)^2}$, i.e. the square root of the sum of the squares of the differences in x and y values of the pair of key points K1(0) and K2(0). The second element of the distance vector **204** is shown as a distance value computed between key point K1(0) (i.e., the nose key point, having index number 0, from the first key point position set **20**) and key point K2(1) (i.e., the neck key point, having index number 1, from the second key point position set **22**). The distance value of this second element is computed as $\sqrt{(x_1' - x_0)^2 + (y_1' - y_0)^2}$. The distance vector **204** thus includes 324 separate elements, each corresponding to a distance value between each pair of (K1,K2) positions, from (K1(0),K2(0)) to (K1(17),K2(17)). For example, the distance vector **204** for image **500** in FIG. **5** would include an element denoted by “(12,3)” corresponding to key point pair (K1(12),K2(3)), i.e., the distance between the left knee key point **512** of the first human body **502** and the right elbow key point **522** of the second human body **504**.

(51) FIG. **7** shows a flowchart of a method **700** for classification of human coactivity in a single image **12** or in a sequence of video frames **52A** through **52B**. The method **700** will be described with reference to system **10**, **50** described above, as implemented by computing system **300**. However, it will be appreciated that the steps of the method **700** may be performed using other means in some embodiments.

(52) Steps **702**, **704**, and **706** are performed for each image of one or more images (e.g., for single image **12**, or for each frame of frames **52A** through **52B**). At **702**, the system (e.g., system **10**) obtains the first key point position set **20** and the second key point position set **22** for the image (e.g., image **12**), as described above. The first key point position set including a key point position for each key point of a plurality of key points of a first human body detected in the image, and the second key point position set including a key point position for each key point of a plurality of key points of a second human body detected in the image. A detailed example implementation of step **702** is described below with reference to FIG. **9**.

(53) At **704**, the encoded representation generation block (e.g., **23** or **53**) processes the first key point position set **20** and the second key point position set **22** to determine a distance between each key point position of the first key point position set **20** and each key point position of the second

key point position set **22**, for example using the distance computation operation **202** described above.

(54) At **706**, the encoded representation generation block (e.g., **23** or **53**) processes the distance between each key point position of the first key point position set and each key point position of the second key point position set to generate an encoded representation of the image (e.g., distance vector **204** included in encoded representation **24**).

(55) After the encoded representations (e.g., distance vectors **204**) have been generated for each image of the one or more images, the method **700** proceeds to step **708**.

(56) At **708**, the encoded representation generation block (e.g., **23** or **53**) provides the encoded representation of each image of the one or more images to a coactivity classifier (e.g., **26** or **56**).

(57) In some examples, optionally, step **708** includes sub-steps **710** and **712** for systems such as second example system **50** configured to process sequences of video frames. At **710**, the encoded representations (e.g., distance vectors **204**) for the plurality of video frames **52A** through **52B** are concatenated or otherwise combined to generate a 3D data volume **254**. At **712**, the 3D data volume **254** is provided as input to the coactivity classifier **56** (e.g., as part of combined encoded representation **54**). The 3D data volume **254** may be provided as a 3D data volume or as a flattened linear vector, as described above.

(58) At **714**, the coactivity classifier (e.g., **26** or **56**) generates classification information **210** classifying a coactivity performed by the first human body (e.g., **502**) and second human body (e.g., **504**) in the one or more images (e.g., image **500**).

(59) FIG. **8** shows steps of a second example method **800** for classifying human coactivity, involving an object, in one or more images. Method **800** includes a number of steps that are identical to those of method **700** and are numbered identically. However, method **800** also includes steps for taking into account one or more object key points in performing the human coactivity classification.

(60) Step **702** is as described above with reference to method **700**. Step **702** is followed by step **802**. At **802**, an object key point position for the image is obtained, such as object key point **542** in image **500**. The object key point position is representative of a position of a key point of an object (e.g., soccer ball **508**) detected in the image.

(61) Step **704** is as described above with reference to method **700**. Step **704** is followed by steps **804** and **806**. At **804**, the first key point position set **20** and the object key point position (e.g., the position of object key point **542**) are processed to determine a distance between each at least one key point position of the first key point position set **20** and the object key point position. At **806**, the second key point position set **22** and the object key point position (e.g., the position of object key point **542**) are processed to determine a distance between each at least one key point position of the second key point position set and the object key point position. In some examples, steps **804** and **806** may be performed by a modified version of the distance computation operation **202** that additionally computes distances from object key point **542** to one or more of the key point positions of the first key point position set **20** and the second key point position set **22**.

(62) Steps **706**, **708**, and **714** are as described above with reference to method **700**. In systems implementing method **800**, the coactivity classifier **26**, **56** is trained using encoded representations that include distance vectors having object key point distance data as generated at steps **804** and **806**.

(63) FIG. **9** shows steps of a detailed example implementation of step **702** of method **700** in which the first key point position set **20** and the second key point position set **22** are obtained using a pose estimation model (e.g., a trained 2D pose estimation CNN used by pose estimation blocks **18**, **19**).

(64) At **902**, the first pose estimation block **18** receives a first bounding box for the first human body **502**, comprising one or more pixel values of a first plurality of pixels of the respective image **500**.

(65) At **904**, the second pose estimation block **19** receives a second bounding box for the second

human body **504**, comprising one or more pixel values of a second plurality of pixels of the respective image **500**.

(66) At **906**, the first pose estimation block **18** processes the first bounding box to generate the key point position for each key point of the plurality of key points of the first human body **502** within the first bounding box (i.e., the first pose estimation block **18** generates the first key point position set **20**).

(67) At **908**, the second pose estimation block **19** processes the second bounding box to generate the key point position for each key point of the plurality of key points of the second human body **504** within the second bounding box (i.e., the second pose estimation block **19** generates the second key point position set **22**).

(68) It will be appreciated that the methods and systems described herein may be applied, in some embodiments, to identification of the coactivity of objects other than human bodies, given appropriate training of the neural networks used to identify key points and the neural networks used to classify coactivity. In further example embodiments, coactivity of anthropomorphic characters or animal bodies could be identified by training the neural networks or other models with suitable training data. In other embodiments, key points could be mapped to moving parts of any system shown in images: for example, vehicles or parts of vehicles.

(69) As described above, some embodiments may process more than one pair of human bodies detected within an image, such as each pair of human bodies detected within the image, to classify coactivities as between multiple pairs of human bodies shown in the image. In some further example embodiments, the described systems and methods may be configured to compute distances between key points of more than two sets of key point positions corresponding to more than two human bodies, and to process the resulting encoded representations to classify coactivities of more than two human bodies. These techniques may, in some examples, be combined with the use of object key points, identification of non-human bodies, and other alternative techniques described above, to classify coactivities of multiple human or non-human objects by means of generating a distance vector and processing said distance vector with a trained neural network or other model according to the principles described herein.

(70) General

(71) The steps and/or operations in the flowcharts and drawings described herein are for purposes of example only. There may be many variations to these steps and/or operations without departing from the teachings of the present disclosure. For instance, the steps may be performed in a differing order, or steps may be added, deleted, or modified.

(72) The coding of software for carrying out the above-described methods described is within the scope of a person of ordinary skill in the art having regard to the present disclosure. Machine-readable code executable by one or more processors of one or more respective devices to perform the above-described method may be stored in a machine-readable medium such as the memory of the data manager. The terms “software” and “firmware” are interchangeable within the present disclosure and comprise any computer program stored in memory for execution by a processor, comprising Random Access Memory (RAM) memory, Read Only Memory (ROM) memory, EPROM memory, electrically EPROM (EEPROM) memory, and non-volatile RAM (NVRAM) memory. The above memory types are examples only, and are thus not limiting as to the types of memory usable for storage of a computer program.

(73) All values and sub-ranges within disclosed ranges are also disclosed. Also, although the systems, devices and processes disclosed and shown herein may comprise a specific plurality of elements, the systems, devices and assemblies may be modified to comprise additional or fewer of such elements. Although several example embodiments are described herein, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the example methods described herein may be modified by substituting, reordering, or adding steps to the disclosed

methods. In addition, numerous specific details are set forth to provide a thorough understanding of the example embodiments described herein. It will, however, be understood by those of ordinary skill in the art that the example embodiments described herein may be practiced without these specific details. Furthermore, well-known methods, procedures, and elements have not been described in detail so as not to obscure the example embodiments described herein. The subject matter described herein intends to cover and embrace all suitable changes in technology.

(74) Although the present disclosure is described at least in part in terms of methods, a person of ordinary skill in the art will understand that the present disclosure is also directed to the various elements for performing at least some of the aspects and features of the described methods, be it by way of hardware, software or a combination thereof. Accordingly, the technical solution of the present disclosure may be embodied in a non-volatile or non-transitory machine-readable medium (e.g., optical disk, flash memory, etc.) having stored thereon executable instructions tangibly stored thereon that enable a processing device to execute examples of the methods disclosed herein.

(75) The term “processor” may comprise any programmable system comprising systems using microprocessors/controllers or nanoprocessors/controllers, digital signal processors (DSPs), application specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs) reduced instruction set circuits (RISCs), logic circuits, and any other circuit or processor capable of executing the functions described herein.

(76) The present disclosure may be embodied in other specific forms without departing from the subject matter of the claims. The described example embodiments are to be considered in all respects as being only illustrative and not restrictive. The present disclosure intends to cover and embrace all suitable changes in technology. The scope of the present disclosure is, therefore, described by the appended claims rather than by the foregoing description. The scope of the claims should not be limited by the embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. A method, comprising: for each image of one or more images: obtaining a first key point position set and a second key point position set for the image; the first key point position set including a key point position for each key point of a plurality of key points of a first human body detected in the image; and the second key point position set including a key point position for each key point of a plurality of key points of a second human body detected in the image; processing the first key point position set and the second key point position set to determine a distance between each key point position of the first key point position set and each key point position of the second key point position set, wherein the distance between each key point position of the first key point position set and all of the key point positions of the second key point position set is determined; and processing the distances between each key point position of the first key point position set and each key point position of the second key point position set to generate an encoded representation of the image; providing the encoded representation of each image of the one or more images to a coactivity classifier that includes a machine learned model that is configured to classify a coactivity of two human bodies; and generating classification information, using the coactivity classifier, classifying a coactivity performed by the first human body and second human body in the one or more images.
2. The method of claim 1, wherein: the one or more images comprises a plurality of frames of a video; and providing the encoded representation of each image of the one or more images to the coactivity classifier comprises: concatenating the encoded representations of the plurality of frames to generate a three-dimensional (3D) data volume; and providing the 3D data volume as input to the coactivity classifier.
3. The method of claim 1, wherein: the one or more images comprises a single image; and the encoded representation of each image of the one or more images comprises a vector of values, each

value being based on the distance between each key point position of the first key point position set and each key point position of the second key point position set.

4. The method of claim 1, wherein: each key point position is a two-dimensional (2D) position within a plane of the respective image; and each distance is a 2D distance within an image plane of the respective image.

5. The method of claim 1, wherein: each key point position is a three-dimensional (3D) position representative of a location of the respective key point in 3D space; and each distance is a 3D distance.

6. The method of claim 1, further comprising: for each image of the one or more images: obtaining an object key point position for the image, the object key point position being representative of a position of a key point of an object detected in the image; processing the first key point position set and the object key point position to determine a distance between each at least one key point position of the first key point position set and the object key point position; and processing the second key point position set and the object key point position to determine a distance between each at least one key point position of the second key point position set and the object key point position; wherein: for each image of the one or more images, the encoded representation of the image is further generated by processing: the distance between each at least one key point position of the first key point position set and the object key point position; and the distance between each at least one key point position of the second key point position set and the object key point position.

7. The method of claim 1, wherein obtaining the first key point position set and the second key point position set for the image comprises: generating the first key point position set and the second key point position set using a pose estimation model configured to: receive a first bounding box for the first human body, comprising one or more pixel values of a first plurality of pixels of the respective image; receive a second bounding box for the second human body, comprising one or more pixel values of a second plurality of pixels of the respective image; process the first bounding box to generate the key point position for each key point of the plurality of key points of the first human body within the first bounding box; and process the second bounding box to generate the key point position for each key point of the plurality of key points of the second human body within the second bounding box.

8. The method of claim 1, wherein the machine learned model comprises a fully-connected neural network.

9. The method of claim 1, wherein: at least one key point position of the first key point position set corresponds to a joint of the first human body; and at least one key point position of the second key point position set corresponds to a joint of the second human body.

10. The method of claim 1, wherein the encoded representation of each respective image comprises a vector comprising a plurality of distance values, the plurality of distance values comprising the distance between each key point position of the first key point position set and each key point position of the second key point position set of the respective image.

11. A system, comprising: a processor; and a non-transitory memory having stored thereon executable instructions that, when executed by the processor, cause the system to: for each image of one or more images: obtain a first key point position set and a second key point position set for the image; the first key point position set including a key point position for each key point of a plurality of key points of a first human body detected in the image; and the second key point position set including a key point position for each key point of a plurality of key points of a second human body detected in the image; process the first key point position set and the second key point position set to determine a distance between each key point position of the first key point position set and each key point position of the second key point position set, wherein the distance between each key point position of the first key point position set and all of the key point positions of the second key point position set is determined; and process the distances between each key point position of the first key point position set and each key point position of the second key point

position set to generate an encoded representation of the image; provide the encoded representation of each image of the one or more images to a coactivity classifier that includes a machine learned model that is configured to classify a coactivity of two human bodies; and generate classification information, using the coactivity classifier, classifying a coactivity performed by the first human body and second human body in the one or more images.

12. The system of claim 11, wherein: the one or more images comprises a plurality of frames of a video; and providing the encoded representation of each image of the one or more images to the coactivity classifier comprises: concatenating the encoded representations of the plurality of frames to generate a three-dimensional (3D) data volume; and providing the 3D data volume as input to the coactivity classifier.

13. The system of claim 11, wherein: the one or more images comprises a single image; and the encoded representation of each image of the one or more images comprises a vector of values, each value being based on the distance between each key point position of the first key point position set and each key point position of the second key point position set.

14. The system of claim 11, wherein: each key point position is a two-dimensional (2D) position within a plane of the respective image; and each distance is a 2D distance within an image plane of the respective image.

15. The system of claim 11, wherein: each key point position is a three-dimensional (3D) position representative of a location of the respective key point in 3D space; and each distance is a 3D distance.

16. The system of claim 11, wherein: the instructions, when executed by the processor, further cause the system to: for each image of the one or more images: obtain an object key point position for the image, the object key point position being representative of a position of a key point of an object detected in the image; process the first key point position set and the object key point position to determine a distance between each at least one key point position of the first key point position set and the object key point position; and process the second key point position set and the object key point position to determine a distance between each at least one key point position of the second key point position set and the object key point position; and for each image of the one or more images, the encoded representation of the image is further generated by processing: the distance between each at least one key point position of the first key point position set and the object key point position; and the distance between each at least one key point position of the second key point position set and the object key point position.

17. The system of claim 11, wherein obtaining the first key point position set and the second key point position set for the image comprises: generating the first key point position set and the second key point position set using a pose estimation model configured to: receive a first bounding box for the first human body, comprising one or more pixel values of a first plurality of pixels of the respective image; receive a second bounding box for the second human body, comprising one or more pixel values of a second plurality of pixels of the respective image; process the first bounding box to generate the key point position for each key point of the plurality of key points of the first human body within the first bounding box; and process the second bounding box to generate the key point position for each key point of the plurality of key points of the second human body within the second bounding box.

18. The system of claim 11, wherein the machine learned model comprises a fully-connected neural network.

19. The system of claim 11, wherein the encoded representation of each respective image comprises a vector comprising a plurality of distance values, the plurality of distance values comprising the distance between each key point position of the first key point position set and each key point position of the second key point position set of the respective image.

20. A non-transitory processor-readable medium containing instructions which, when executed by a processor of a system, cause the system to: for each image of one or more images: obtain a first key

point position set and a second key point position set for the image; the first key point position set including a key point position for each key point of a plurality of key points of a first human body detected in the image; and the second key point position set including a key point position for each key point of a plurality of key points of a second human body detected in the image; process the first key point position set and the second key point position set to determine a distance between each key point position of the first key point position set and each key point position of the second key point position set, wherein the distance between each key point position of the first key point position set and all of the key point positions of the second key point position set is determined; and process the distances between each key point position of the first key point position set and each key point position of the second key point position set to generate an encoded representation of the image; provide the encoded representation of each image of the one or more images to a coactivity classifier that includes a machine learned model that is configured to classify a coactivity of two human bodies; and generate classification information, using the coactivity classifier, classifying a coactivity performed by the first human body and second human body in the one or more images.
